

Improving operational flexibility by regulating extraction steam of high-pressure heaters on a 660 MW supercritical coal-fired power plant: A dynamic simulation

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HIGHLIGHTS

- Dynamic models of a 660 MW supercritical coal-fired power plant are developed.
- Five measures of regulating extraction steam of high-pressure heaters are analyzed.
- Dynamic responses of main thermodynamic parameters and output power are described.
- The best strategy of operational flexibility on coal-fired power plant is achieved.

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Extraction steam of high-pressure heater
Coal-fired power plant
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ABSTRACT

The operational flexibility of a thermal power plant has played an essential and promising role in accommodating the increment of variability in the supply and demand sides in China. The rapid activation of thermal storage in entire coal-fired power plant lies at the heart for the strategy. Understanding the thermodynamic characteristics of the power unit in the transient process quantitatively remains a challenge. In this study, dynamic simulations of an entire 660 MW supercritical coal-fired power plant were conducted via GSE software, and the models were validated in the steady state and transient processes. Then, five different measures were introduced to regulate the extraction steam of high-pressure heaters for operational flexibility. The dynamic characteristics of the main thermodynamic parameters and output power were described and compared. Moreover, the operational flexibility of these measures was discussed. It turns out that: among the five measures, the change degrees of pressure, flowrate, and temperature in main devices increase with the increment in the number of throttled valves and/or the degree of feedwater bypass. The feedwater temperature at steady state reduces by 94.6 °C at most, and the maximum temperature change rate of metal slabs in HP heaters is $-44.6\text{ }^{\circ}\text{C min}^{-1}$. The dynamic process for output power under different measures has two different ramp stages, namely, a rapid stage and a slow stage. Furthermore, compared with other measures, 100% throttling extraction steam of #1, #2, and #3 HP heaters has the best operational flexibility, that is, the maximum average power ramp rate in a minute, power capacity, and energy capacity are 6.19% of rate power per minute, 48.40 MW, 5.58 MW h, respectively. The average power ramp rate, power capacity and energy capacity increase with the increment in the number of throttled valves and/or the degree of feedwater bypass. This work is expected to provide a detailed reference on the use of turbine thermal storage to improve the operational flexibility of coal-fired power plants.

1. Introduction

Recently, the use of renewable energy sources (RES), such as solar, wind, wave, and tidal energies, has dramatically increased around the world [1–3]. However, increasing the share of intermittent RES in power system poses challenges in terms of increasing net load

variability and maintaining grid stability and security [4–6]. Therefore, the assessment and planning of power system flexibility has gradually attracted researchers' attention [7–9]. Chronologically classified, flexible resources include super short-term, short-term, mid-term, and long-term categories [10].

Operational flexibility, which is the technical ability of a power unit

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Nomenclature		hphs	HP heater subsystem
Abbreviations		Symbols	
ECO	economizer	A	cross-sectional area (m^2)
FWB	feedwater bypass	C_p	heat capacity ($\text{kJ kg}^{-1} \text{K}^{-1}$)
BFPT	boiler feedwater pump turbine	F	flowrate (kg s^{-1})
HP	high-pressure	f	friction factor (MPa m^{-1})
IP	intermediate-pressure	g	gravitational acceleration (m s^{-2})
LHV	lower heating value	h	enthalpy (kJ kg^{-1})
LP	low-pressure	K	turbine power constant
RES	renewable energy sources	P	pressure (MPa)
RH	reheater	P_o	initial output power (MW)
SH	superheater	P_{max}	maximum output power (MW)
TES	throttling extraction steam	$P(t)$	real-time output power (MW)
THA	turbine heat acceptance	$\dot{Q}$	heat transfer flow (kJ s^{-1})
Subscript/superscript		Q_s	thermal storage (MJ)
ar	as-received basis	q_s	thermal storage rate (MW)
f	fluid	S	flow source terms (kg s^{-1})
in	inlet	T	temperature (K)
out	outlet	t	time (s)
ext	extraction	v	velocity of fluid (m s^{-1})
s	slab	W	shaft power (kW)
rad	radiation	X	thickness (m)
g	gas	Z	space flow direction of fluid
l	liquid	Greek symbols	
sat	saturation	Γ	phase transition term (kg s^{-1})
src	source	α	mass fraction in the node (%)
f-w	fluid to wall	ρ	density (kg m^{-3})
f-f	fluid to fluid	μ	power ramp rate (MW min^{-1})
c	convection	π	power capacity (MW)
ave	average	ε	energy capacity (MJ)
bs	boiler system	δ	source flow per unit length (m^{-1})

to modulate electrical power feed-into the grid and/or power feed-out from the grid over time, plays a crucial role in the transition of current power systems [11]. Various alternatives can be used to offer operational flexibility to the grid, that is, on the supply side, on the demand side and energy storage system [12]. On the demand side, demand side management provides ancillary service to the electric grid by modifying the electricity demand of customers [13]. However, the load is random and the demand response curve is uncertain. Thus, operational flexibility is difficult to achieve on the demand side. Although energy storage technologies based on the form of stored energy have different kinds [14], and ancillary services, such as primary and secondary frequency control reserves, can be provided by energy storage devices [15] and renewable energy sources [16], the technologies of energy storage units are immature and limited at present. Fortunately, on the supply side, conventional power plants can dynamically respond to outlined system disturbances and the processes can be fully controllable, thus providing a good choice for operational flexibility. Meanwhile, as a vital component of mature and reliable electricity markets, coal-fired power plants account for more than 60% of the power infrastructure in China [17], and the power units are the main participants in the load cycling and frequency regulation. For the foreseeable future, coal-fired power plants will have to operate alongside new renewables and play an essential role in accommodating the increasing variability on the supply side [18]. Therefore, operational flexibility of coal-fired power plants should be provided with more importance and studied profoundly.

For coal-fired power plants, operational flexibility is the rapid power response capability for frequency regulation, which usually

demand the activation of thermal storage in power units. Thus, boiler-turbine coordinated control, which mainly uses the thermal storage in the boiler system, is a popular strategy to regulate steam turbine output power in thermal power plants [19]. However, the slow response is caused by the large delay from mass transport, fuel grinding, and heat transfer to the water-steam cycle [20]. Therefore, the most studies have focused on optimal of control systems. Sondhi and Hote [21] designed a fractional order PID controller for single-area LFC for all three types of turbines, namely, non-reheated, reheated, and hydro turbines [22]. However, to optimize the production process and design a related control system, an appropriate process dynamic model should be built first. Additionally, the development of industries and technologies related in the gradual takeover of supercritical even ultra-supercritical thermal power plants in China's electricity market, thereby promoting the economy of thermal power systems. However, the thermal storage in the once-through boiler decreases with the promotion. This contradiction motivates the development and maturity of other utilization strategies of thermal storage in thermal power plants.

The dynamic simulation of a conventional power plant is an essential step to achieve the desirable thermodynamic performance under various operational conditions. Gao and Dai [23] built a linear model for the dynamic analysis of a power system. Starkloff et al. [22] developed a full-scale dynamic model of coal-fired power plant via Advanced Process Simulation Software, and the increased flexibility of the validated model in large, conventional coal-fired power plants was highly relevant for further investigations. Alobaid et al. [24,25] proposed a comprehensive review regarding the dynamic simulation of various thermal power plants, such as coal-fired power plant,

combined-cycle power plant, nuclear power plant and concentrated solar power plant [26]. Sedic et al. [27] developed the nonlinear dynamic models of a natural water circulation boiler using the basic laws of conservation of mass, energy, and momentum via MATLAB/Simulink software. Wang et al. [28] established the dynamic models of a 660 MW supercritical unit via GSE software, and the energy consumption of the power plant was obtained at different cycling rates and compared with the results calculated using steady-state models. Fan et al. [26] established a precise dynamic mathematical model of an ultra-supercritical OTB unit under dry operating conditions [29]. Liu et al. [30] established a lumped parameter dynamic model that with relatively low complexity while capturing the essential overall importance of the plant. Wang et al. [31] established a dynamic mathematical model of 300 MW thermal power plant based on the principles of mass and energy balance, which provided reference for the simulation and control during the rapid cutback of thermal power units. Benato et al. [29,32] proposed a complete dynamic model of combined cycle power plants via Dymola and MATLAB/Simulink software, which could simulate the trends of the main thermodynamic parameters during various transient operating conditions.

Apart from the stored energy of the boiler, a large amount of thermal storage in the turbine side exists. To use the thermal storage in the deaerator and the low pressure (LP) heaters of the turbine subsystem, Lausterer [33] proposed a measure called condensate throttling, which could shortly and rapidly increase the power output of thermal power plants. Hu et al. [34] proposed a non-linear dynamic model of condensate throttling system using compound modeling method. The results showed that the responses of the established model strongly correlated with the responses of the actual unit. Wang et al. [35,36] set up the static and dynamic models for the condensate throttling of a 1000 MW coal-fired power unit and developed an improved coordinated control strategy that combines with condensate throttling. The results revealed that the control performance of the improved method was markedly superior to the traditional control strategy. Using Dymola software, Huebel et al. [20,37,38] established dynamic simulation models on a lignite power plant. Moreover, the four ways of providing additional power output were compared and indicated that throttling steam supply of high pressure (HP) preheaters could be used to enhance the primary control reserves over the load range of the power plant.

In summary, the previous studies focused on the assessment for the operational flexibility of energy storage technologies and renewable energy sources. Although conventional power plants were the main participants in the load cycling and frequency regulation in China,

assessment for the operational flexibility of coal-fired power plant was scarce. Moreover, researchers mainly focused on the use of thermal storage in LP heaters for operational flexibility, and some analyses were conducted. By contrast, the regulation of the extraction steam of HP heaters could activate more thermal storage, and the potential for operational flexibility could be considerable, but detailed models and quantitative discussion were scarce. Furthermore, although some dynamic models exist in thermal power plants, the thermodynamic characteristics, which depict the nature of the complicated power unit in transient process, were not considered sufficiently. Accordingly, to the best of our knowledge, no relevant studies focused on the operational flexibility of the thermal power plant based on the mechanism dynamic models.

In this study, the dynamic simulation models of coal-fired power plants were developed via the GSE software. Furthermore, five different measures for regulating the extraction steam of HP heaters were introduced to activate the thermal storages in HP heaters subsystem and boiler system. By using these measures, the dynamic characteristics of the main thermodynamic parameters and output power were described and compared, and the evaluations of operational flexibility were discussed. This study mainly aims to exploit the potential use of turbine thermal storage, and the performance parameters and strategies obtained in our study can provide a detailed reference to improve the operational flexibility of coal-fired power plants.

This study is organized as follows. Section 2 describes the reference case and simulation details and implements model verifications. Section 3 defines the performance indexes of thermal storage and operational flexibility. Section 4 presents the main results and discussion. Section 5 provides the concluding remarks.

2. Modeling and simulation details

2.1. Reference case

In our study, a 660 MW supercritical coal-fired power plant was selected. Fig. 1 illustrates the thermal system. The total power system has two main subsystems: boiler and turbine. At the boiler subsystem, the first inlet fluid is the feedwater that comes from the outlet of #1 HP heater. Subsequently, the feedwater is successively heated by binary-stage economizers, water walls, and triple-stage superheaters to produce the live steam with parameters of 25.40 MPa and 571 °C at rated condition. The second inlet fluid is the cold reheat steam from the outlet of the HP turbine. Subsequently, the steam is heated by binary-stage reheaters to produce the reheat steam with parameters of 4.42 MPa and

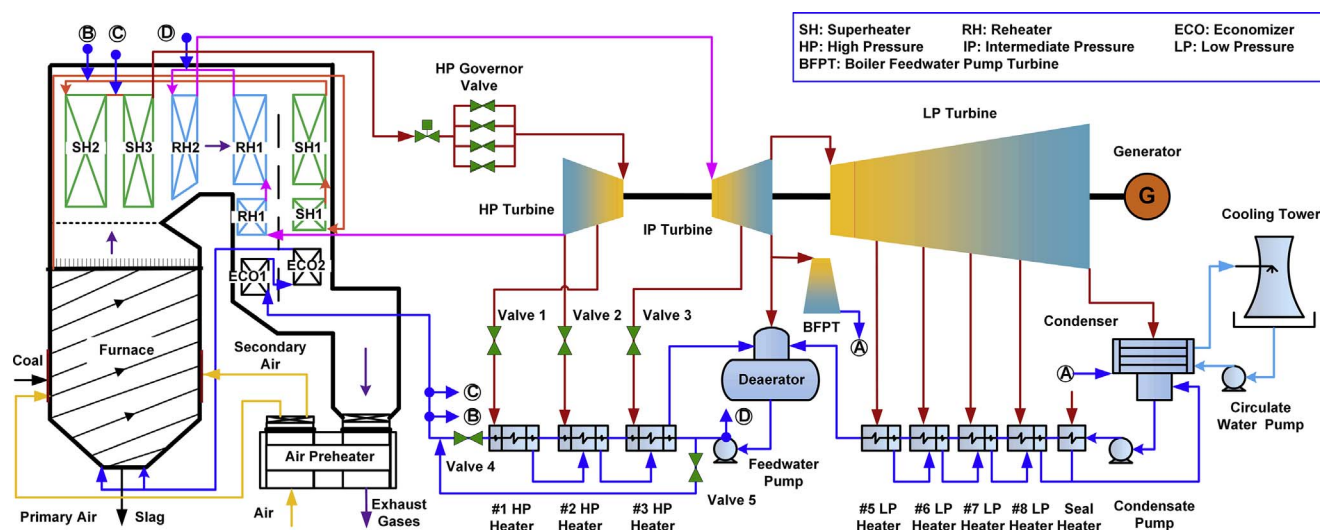


Fig. 1. Schematic thermal system of supercritical coal-fired power plant.

569 °C at rated condition. The bituminous coal is designed, and the parameters are listed in Table 1. Air is preheated in the air preheater and then divided into two parts, namely, the primary hot air and the secondary hot air. Finally, the pre-heated air enters the boiler.

At the turbine subsystem, the inlet parameters at rated condition are 24.20 MPa, 566 °C, and 566 °C for the live steam pressure, live steam temperature, and reheat steam temperature, respectively. The four turbines, namely, HP, IP, LP, and boiler feedwater pump turbines are used to produce power. The heat regenerative system contains three HP heaters, four LP heaters, a deaerator, and other heaters. The wet-cold end system includes a condenser, a cooling tower, several circulate water pumps, and other auxiliary devices. In our study, the valves at the extraction steam pipelines of #1, #2, and #3 HP heaters are called valve 1, valve 2, and valve 3, respectively. The valves at the feedwater and feedwater bypass pipeline are called valve 4 and valve 5, respectively. Those valves will be used for our subsequent regulating strategies.

2.2. Mathematical models

Building accurate dynamic models that can simulate the flow and heat transfer in two-phase fluids is the key in studying the thermodynamic characteristics of a power plant. The GSE software was used in this study, and Fig. 2 shows the simplified description of GSE units. Several units, such as node, slab, pump, valve, flow link, and boundary, are based on the GSE JTOPMERET module. For every node, geometric parameters, including hydraulic diameter, cross-sectional area, thickness, and length of the pipe, are the input data. Moreover, these parameters begin from the fundamental conservation equations of mass, momentum, and energy to generate a set of programs that simulates two-phase flow and temperature. Furthermore, by separating the mass, momentum, and energy balance of the gas and liquid phases, more accurate results for non-equilibrium phenomenon are obtained [39]. The main governing equations are described as follows.

2.2.1. Conservation of mass

$$A \frac{\partial(\alpha \rho)_f}{\partial t} + \frac{\partial F_f}{\partial Z} = \sum \frac{\partial S_f}{\partial Z} + \frac{\partial \Gamma_f}{\partial Z} \quad (1)$$

where A is the cross-sectional area of the node, m^2 ; α_f is the fluid fraction in the node, %; ρ_f is the fluid density in the node, kg m^{-3} ; t is the time, s ; F_f is the flowrate of fluid, kg s^{-1} ; Z is the space flow direction of fluid; S_f is the flow source terms of fluid, kg s^{-1} , and Γ_f is the phase transition term of fluid, kg s^{-1} .

2.2.2. Conservation of momentum

$$\frac{dF_f}{dt} = -\frac{\alpha_f A}{K_u} \left(\frac{dP}{dZ} + f_{f-w} + f_{f-f} + K_u \rho_f g - \delta P_{\text{pump}} \right) + \sum \delta S_f v_{\text{src}} \quad (2)$$

where P is the node pressure, MPa; f_{f-w} is the friction factor of fluid to wall, MPa m^{-1} ; f_{f-f} is the friction factor of fluid to fluid, MPa m^{-1} ; K_u is the unit conversion factor, $\text{MPa m s}^2/\text{kg}$; g is the fluid gravity, m s^{-2} ; δP_{pump} is the differential of pump pressure at per unit length, MPa m^{-1} ; δ is the source flow at per unit length, m^{-1} , and v_{src} is the flow velocity of source fluid, m s^{-1} .

2.2.3. Conservation of energy

$$A \frac{\partial(\alpha \rho h)_f}{\partial t} + \frac{\partial(Fh)_f}{\partial Z} = A(\Gamma h_{\text{sat}} + \dot{Q} - W)_f + \sum \delta S_f h_{\text{src}} \quad (3)$$

where h_f is the enthalpy of the fluid, kJ kg^{-1} ; h_{sat} is the saturation enthalpy of the fluid, kJ kg^{-1} ; $\dot{Q}$ is the heat transfer term which contains several parts, including heat transfer to the other phase, heat transfer to a heat slab, and external heat sources, kJ s^{-1} ; W is the shaft power calculated in the turbine nodes, kJ s^{-1} , and h_{src} is the enthalpy of the flow source term, kJ kg^{-1} .

2.2.4. Heat slab energy conservation equation

For every heat slab, the real characteristic parameters of metal tubes, such as material type, thickness, surface area, density, and heat capacity of metal, are inputted in the unit. Then, the energy conservation equation is as follows:

$$(A \rho X C_p)_s \frac{dT_s}{dt} = \sum \dot{Q}_{c,l,\text{in}} + \sum \dot{Q}_{c,g,\text{in}} + \sum \dot{Q}_{\text{rad}} - \sum \dot{Q}_{c,l,\text{out}} - \sum \dot{Q}_{c,g,\text{out}} \quad (4)$$

where A_s is the surface area of the heat slab, m^2 ; ρ_s is the density of the heat slab, kg m^{-3} ; X_s is the thickness of the heat slab, m ; C_{p_s} is the heat capacity of the heat slab, $\text{kJ kg}^{-1} \text{K}^{-1}$; T_s is the temperature of the heat slab, °C; $\dot{Q}_{c,l,\text{in}}$ is the liquid convection heat flow at the slab inlet, kJ s^{-1} ; $\dot{Q}_{c,g,\text{in}}$ is the gas convection heat flow at the slab inlet, kJ s^{-1} ; $\dot{Q}_{\text{rad}}$ is the inlet radiation heat flow at the slab inlet, kJ s^{-1} ; $\dot{Q}_{c,l,\text{out}}$ is the liquid convection heat flow at the slab outlet, kJ s^{-1} , and $\dot{Q}_{c,g,\text{out}}$ is the gas convection heat flow at the slab outlet, kJ s^{-1} .

2.2.5. Turbine shaft work equation

In the turbine model, the pressure nodes are used to represent the inlet, exhaust, and all extraction points of the turbine. The flow and node associated with each extraction and/or exhaust are referred as stages. The model calculates the shaft work conducted on or by the blades of the stages based on the thermodynamic properties at each stage inlet and exhaust. The result are suitable for the appropriate thermodynamic conditions at each stage and accurate dynamic responses to the transient conditions around the system. The equation of shaft work in turbine is as follows:

$$W = K_{t1} \sqrt{\frac{F_g}{\rho_{\text{in}}}} (P_{\text{in}} - P_{\text{out}})^{K_{t2}} \quad (5)$$

where K_{t1} and K_{t2} are the turbine power constants, which can be calculated according the heat balance data; F_g is the gas flowrate of the node, kg s^{-1} ; ρ_{in} is the gas density of the stage inlet, kg m^{-3} , and P_{in} and P_{out} are the pressure of stage inlet and outlet, respectively, MPa.

In this study, pressure, flow, and heat flux boundaries are used to control the boundary conditions. Moreover, in the Supporting Information, Section 1 introduces the calculation models between pressure and flowrate, Section 2 illustrates the calculation models and characteristic curves of the valves, and Section 3 lists the geometric parameters of the extraction steam pipelines. The correlations of heat transfer coefficient and the details of the numerical solution, assumptions, and restrictions of these dynamic models are presented in the previous study by our group [28].

2.3. Simulation models

The boiler-turbine coupled models were developed and depicted in Fig. 3. In this study, the nodes are used to simulate the space of fluids in different devices and pipelines, whereas the slabs are used to simulate the heat transfer metal wall. Therefore, fluid and metal storages are under our consideration. The coal burning process in the furnace is self programmed using Fortran language and added in the software. To study the dynamic properties of regulating the extraction steam of HP heaters, the following hypotheses are used in the transient process:

- (1) The type and quantity of coal are fixed.

Table 1
Ultimate and proximate analysis of bituminous coal (as-received basis).

Items	Ultimate analysis/wt%					Proximate analysis/wt%				LHV/ (kJ kg ⁻¹)
	C _{ar}	H _{ar}	O _{ar}	N _{ar}	S _{ar}	M _{ar}	A _{ar}	V _{ar}	FC _{ar}	
Data	52.5	3.03	9.96	0.54	0.64	16.01	17.32	37.66	29.01	19652

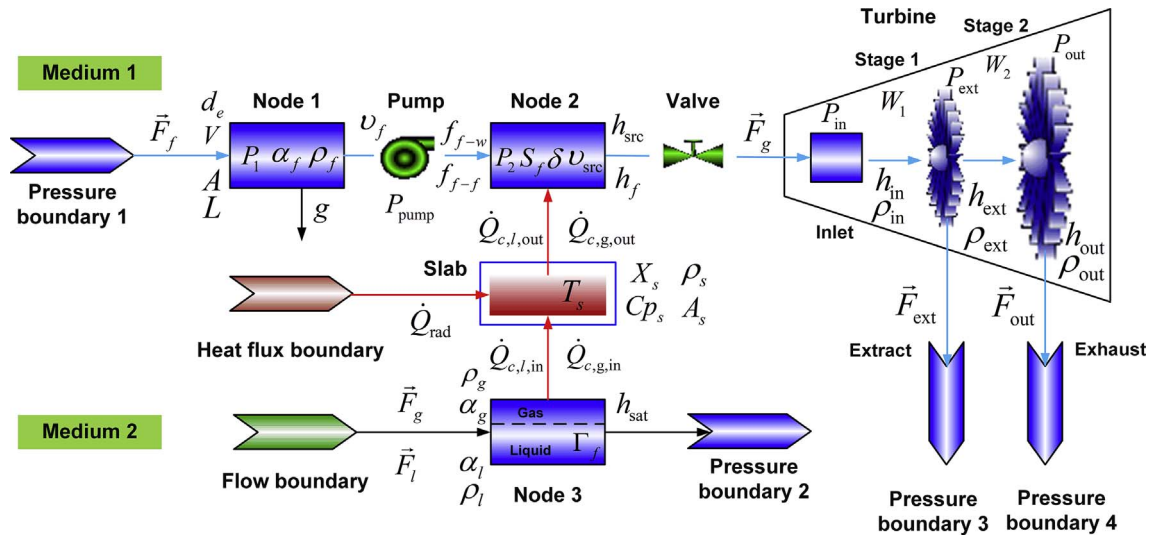


Fig. 2. Simplified description of GSE JTOPMERET units.

- (2) The flowrate of feedwater that originated from the deaerator is fixed.
- (3) The inlet flowrate and temperature of air are fixed.
- (4) The inlet flowrate and temperature of circulating water are fixed.
- (5) The rapid cut-off valves and electric control valves are used in extraction steam pipelines and feedwater pipelines, respectively.

2.4. Model verification

2.4.1. Validation of models in the steady state

In this study, the simulated values and designed data were compared based on the main parameters of the boiler in the steady state at three different working conditions. The results are illustrated in Table 2. It turns out that the simulated values of the main parameters are consistent with the designed data, and the maximum relative error is 0.78%, thereby validating the precision of our models in the steady state.

2.4.2. Validation of models in the transient process

A reliable software called GSE, which is a high-fidelity and high-definition dynamic simulator for coal-fired power plant, was selected for our study. In the models, geometric parameters, including hydraulic diameter, cross-sectional area, thickness, length and material type at the different type of devices and pipelines, are the inputted data for the nodes and slabs, which can guarantee that all the properties of devices and pipelines in thermal power units are under our consideration. However, in power station, total power systems are coupled and comprise several control systems, and the accuracy and acquisition frequency of online measurement for the main parameter are very insufficient. Thus, the full-scope test data in the live are very difficult to verify our simulated models.

Nevertheless, an experiment was designed by our group [28] to investigate the dynamic characteristic of a heat exchanger. The details are described in Section 4 of Supporting Information. The result shows that the simulated and experimental results show a good agreement. Moreover, the thermal power plant is composed of many different heaters, and the regenerative heaters are among the most important

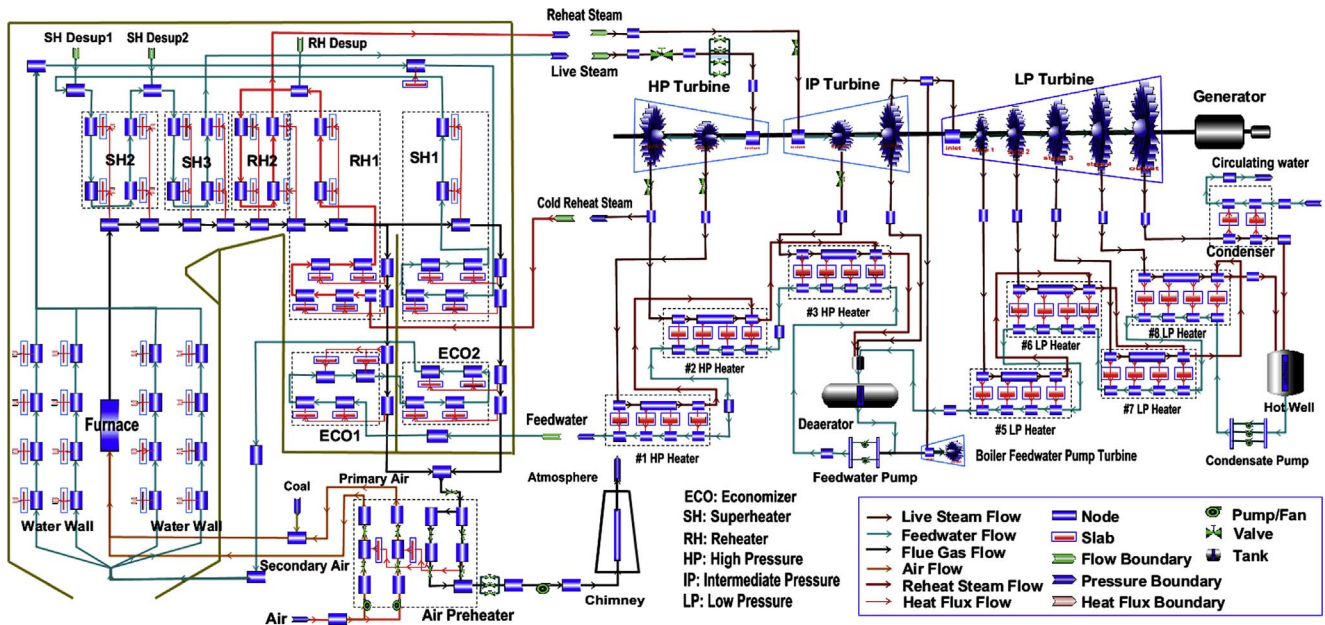


Fig. 3. Simulated models of boiler-turbine coupled system.

Table 2

Comparison of simulated value with designed data based on the main parameters of boiler.

Main parameters	100% THA			75% THA			50% THA		
	Designed Data	Simulated result	Relative errors (%)	Designed data	Simulated result	Relative errors (%)	Designed data	Simulated result	Relative errors (%)
Live steam temperature (°C)	571.0	570.4	0.09%	571.0	571.1	−0.02%	571.0	571.7	−0.13%
Live steam pressure (MPa)	25.29	25.29	0.00%	24.74	24.74	0.00%	17.30	17.30	0.00%
Feedwater temperature (°C)	278.0	277.9	0.05%	259.0	258.9	0.05%	235.0	235.0	0.01%
Feedwater pressure (MPa)	28.62	28.81	−0.68%	26.55	26.56	−0.03%	18.42	18.28	0.78%
Reheat steam inlet temperature (°C)	315.0	315.4	−0.12%	294.0	294.1	−0.03%	292.0	292.2	−0.06%
Reheat steam inlet pressure (MPa)	4.37	4.38	−0.21%	3.26	3.26	0.00%	2.13	2.12	0.40%
Reheat steam outlet temperature (°C)	569.0	569.1	−0.01%	569.0	569.0	0.01%	569.0	569.5	−0.08%
Reheat steam inlet pressure (MPa)	4.18	4.18	0.00%	3.11	3.11	0.00%	2.02	2.02	0.00%
Exhaust gas temperature (°C)	130.0	129.8	0.15%	118.0	117.7	0.27%	108.0	108.6	−0.56%

because two-phase heat transfer is involved. Therefore, HP regenerative heater system including #1, #2, and #3 HP heaters (shown in Fig. 4) was selected as an example. The flowrate and enthalpy of every extraction steam inlet, and the inlet flowrate and enthalpy of feedwater at #3 HP heater are used as input data for our models in the load-up cycling. Subsequently, our simulated results via GSE software and the test data were compared and displayed in Fig. 5. It turns out that the flowrate and temperature of feedwater at #1 HP heater outlet and flowrate and temperature of drain at #3 HP heater outlet increased

during the load-up cycling process, and the maximum relative errors of feedwater and drain temperature are 0.89% and 1.86%, respectively. Overall, our simulated results are in agreement with the test data, thereby validating the precision of our dynamic models.

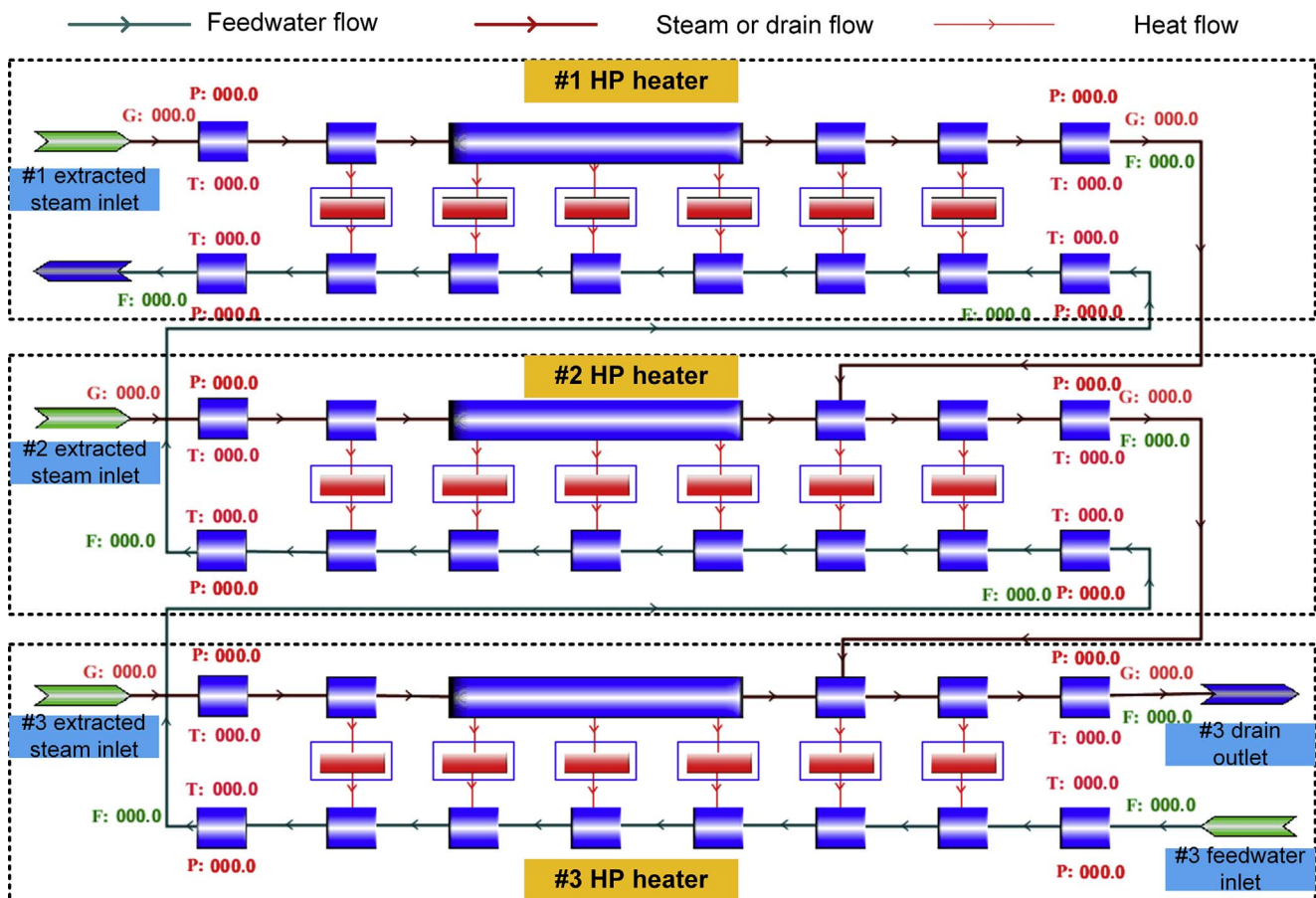


Fig. 4. Simulated models of HP regenerative heater system, including #1, #2, and #3 HP heaters.

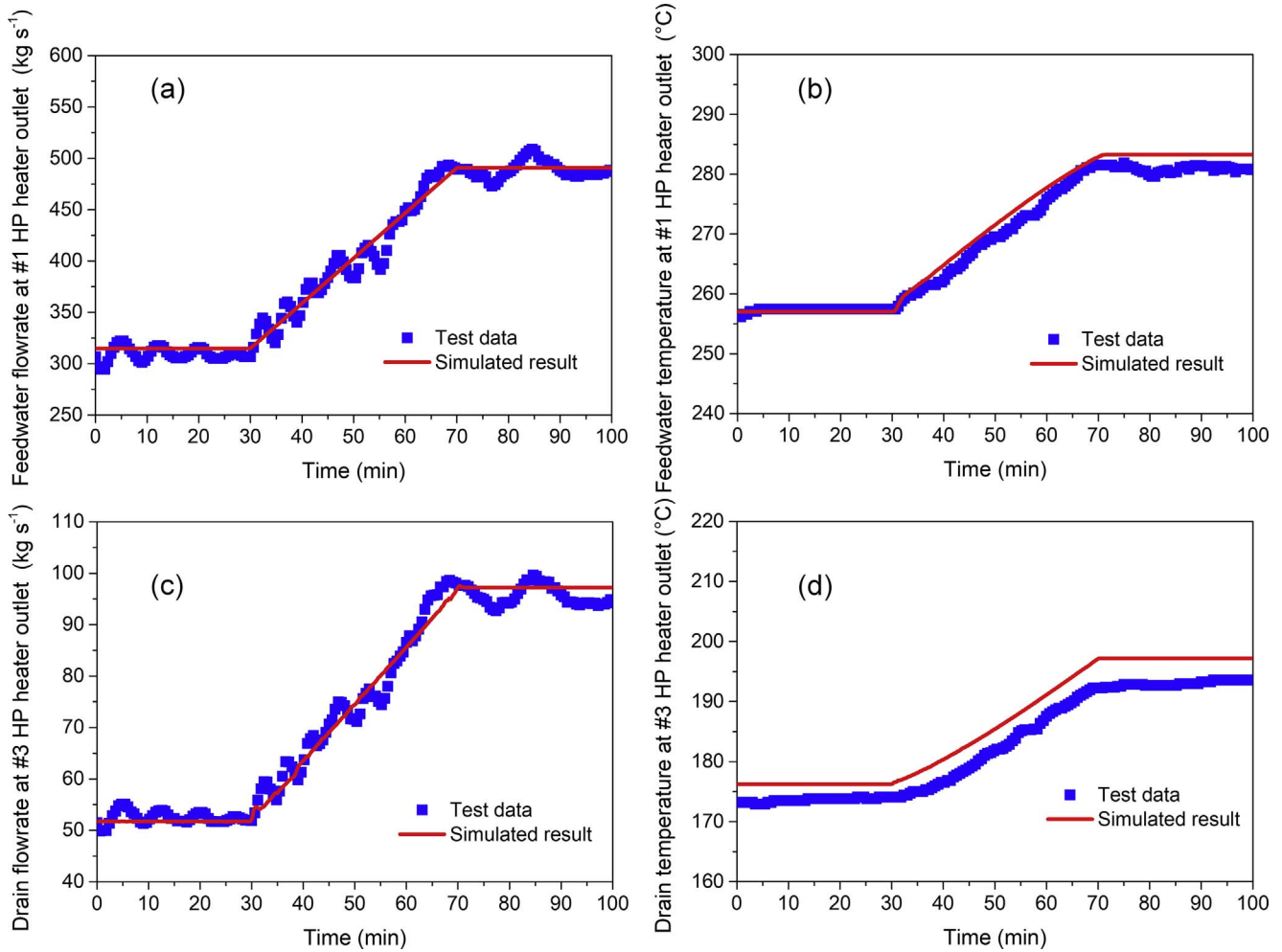


Fig. 5. Parameter changes of HP regenerative heater system in load-up cycling process. (a) Feedwater flowrate at #1 HP heater outlet; (b) feedwater temperature at #1 HP heater outlet; (c) drain flowrate at #3 HP heater outlet; (d) drain temperature at #3 HP heater outlet.

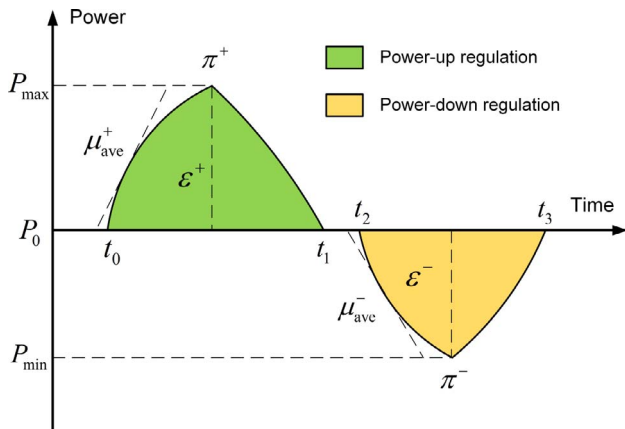


Fig. 6. Description of flexibility metrics of thermal power plant during operation.

Table 3
Details of simulated measures that can regulate extraction steam of HP heaters.

Measures	Description	Implementing measures	Required time for the regulation (s)
TES 1	100% throttling extraction steam of #1 HP heater	Valve 1 cutting-off	1
TES 2	100% throttling extraction steam of #1 and #2 HP heaters	Valve 1 and valve 2 cutting-off	1
TES 3	100% throttling extraction steam of #1, #2, and #3 HP heaters	Valve 1, valve 2, and valve 3 cutting-off	1
FWB 1	50% feedwater bypass	Coordinating by valve 4 and valve 5	30
FWB 2	100% feedwater bypass	Valve 4 cutting off, valve 5 opening	60

3. Performance indexes of thermal storage utilization and operational flexibility

3.1. Thermal storage utilization indexes

For the specific supercritical power unit, the entire system consists of many pipelines and heating devices, which contains hundreds of cubic meters of water in liquid or vapor form, and hundreds of tons of metal slabs at high temperature levels are also available. In this study, the thermal storages of fluids and metal slabs are defined as follows:

$$Q_{s,fluid} = m_{fluid} \Delta h_{fluid} \text{ [kJ]} \quad (6)$$

$$Q_{s,slab} = m_{slab} c_{p,slab} \Delta T_{slab} \text{ [kJ]} \quad (7)$$

where m_{fluid} and m_{slab} are the masses of the fluid node and metal slab, respectively, kg; Δh_{fluid} is the enthalpy difference of the fluid node, kJ kg^{-1} , and ΔT_{slab} is the temperature difference, K.

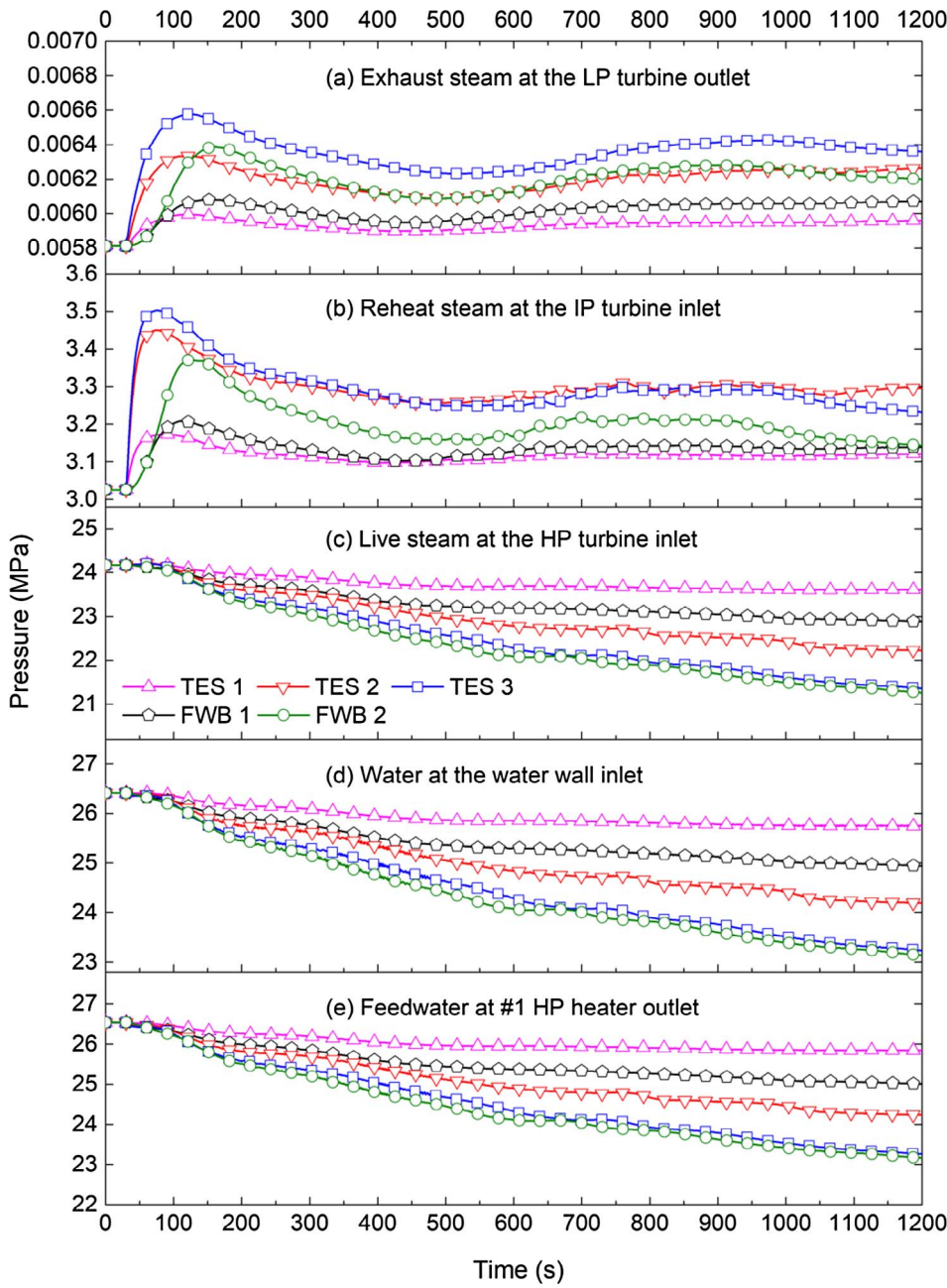


Fig. 7. Pressure dynamic responses of key parameters under five different measures.

The thermal storages of the boiler system and HP heater subsystem will be mainly activated by regulation of the extraction steam of HP heaters. Thus, the total thermal storages of these systems are defined as follows:

$$Q_s = Q_{s,fluid}^{bs} + Q_{s,slab}^{bs} + Q_{s,fluid}^{hphs} + Q_{s,slab}^{hphs} \text{ [MJ]} \quad (8)$$

where $Q_{s,fluid}^{bs}$ and $Q_{s,slab}^{bs}$ are the thermal storages of total fluid nodes and metal slabs in the boiler system, respectively, MJ; $Q_{s,fluid}^{hphs}$ and $Q_{s,slab}^{hphs}$ are the thermal storages of total fluid nodes and metal slabs in the HP heater subsystem, respectively, MJ. Moreover, the thermal storage rate is defined as:

$$q_s = \frac{dQ_s}{dt} \text{ [MW]} \quad (9)$$

3.2. Metrics for assessing operational flexibility

Three metrics, namely, power ramp rate, power capacity and energy

capacity, can be used to assess the operational flexibility of the thermal power system [11], as shown in Fig. 6. These metrics are defined as follows:

Power ramp rate:

$$\mu = \frac{dP(t)}{dt} \text{ [MW min}^{-1}\text{]} \quad (10)$$

Power capacity:

$$\pi = P_{\max} - P_0 \text{ [MW]} \quad (11)$$

Energy capacity:

$$\varepsilon = \int_{t_0}^{t_1} (P(t) - P_0) dt \text{ [MJ]} \quad (12)$$

where $P(t)$, $P_{\max}$, and P_0 are the real-time, maximum, and initial output power, respectively, MW.

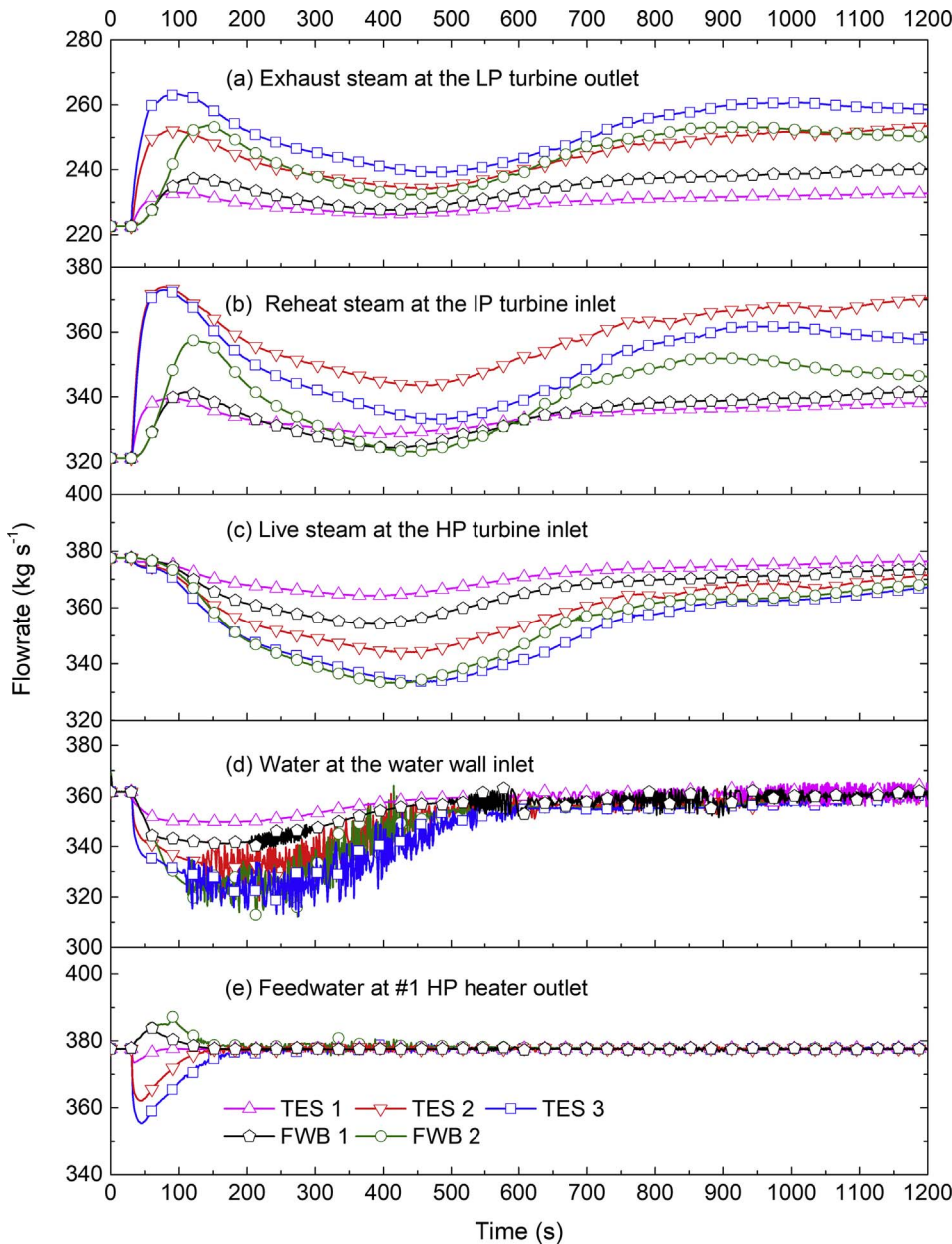


Fig. 8. Flowrate dynamic responses of key parameters under five different measures.

4. Results and discussion

The regulation of the extraction steam of HP heaters has several measures. One is by directly throttling extraction steam (TES) at the extraction steam pipelines, and the other is by indirect feedwater bypass (FWB). The dynamic characteristics under the following five measures, which were all conducted at 30 s, were investigated based on the 75% THA working condition. The details are reported in Table 3.

4.1. Dynamic responses of thermodynamic parameters

The dynamic responses of the thermodynamic parameters, which mainly reflect the thermodynamic state of the key parameters, were obtained and analyzed. Furthermore, a solid foundation for the economy and thermal stress analyses of the power system were provided.

4.1.1. Dynamic responses of pressure

Fig. 7 shows the pressure dynamic responses of the key parameters

under five different measures. It can be seen that when the different measures are performed, the pressures of reheat steam at the IP turbine inlet and exhaust steam increase sharply to reach an extremum and then decrease slightly until they finally reach a steady state. Meanwhile, the response rates of these parameter via TES measures are quicker than those via FWB measures, which is ascribed to that the regulating times through TES measures are shorter than those via FWB measures. For the live steam pressure at the HP turbine inlet, water pressure at the water wall inlet, and feedwater pressure at #1 heater outlet, these trends coincide, that is, all of them decrease and then reach a steady state gradually. However, the response rates of these parameters via TES measures are slower than those via FWB measures because of the time delay of the heat transfer in the HP heater system. Moreover, among the three TES measures, the change degrees of pressure increase with the increment in the number of throttled valves. Moreover, between the two FWB measures, the change degrees of pressure increase with the rise in the degree of feedwater bypass. Furthermore, on the steady state, the pressures of live steam at the HP turbine inlet, water at the water wall inlet, and feedwater at #1 heater outlet via TES 3 (100% throttling

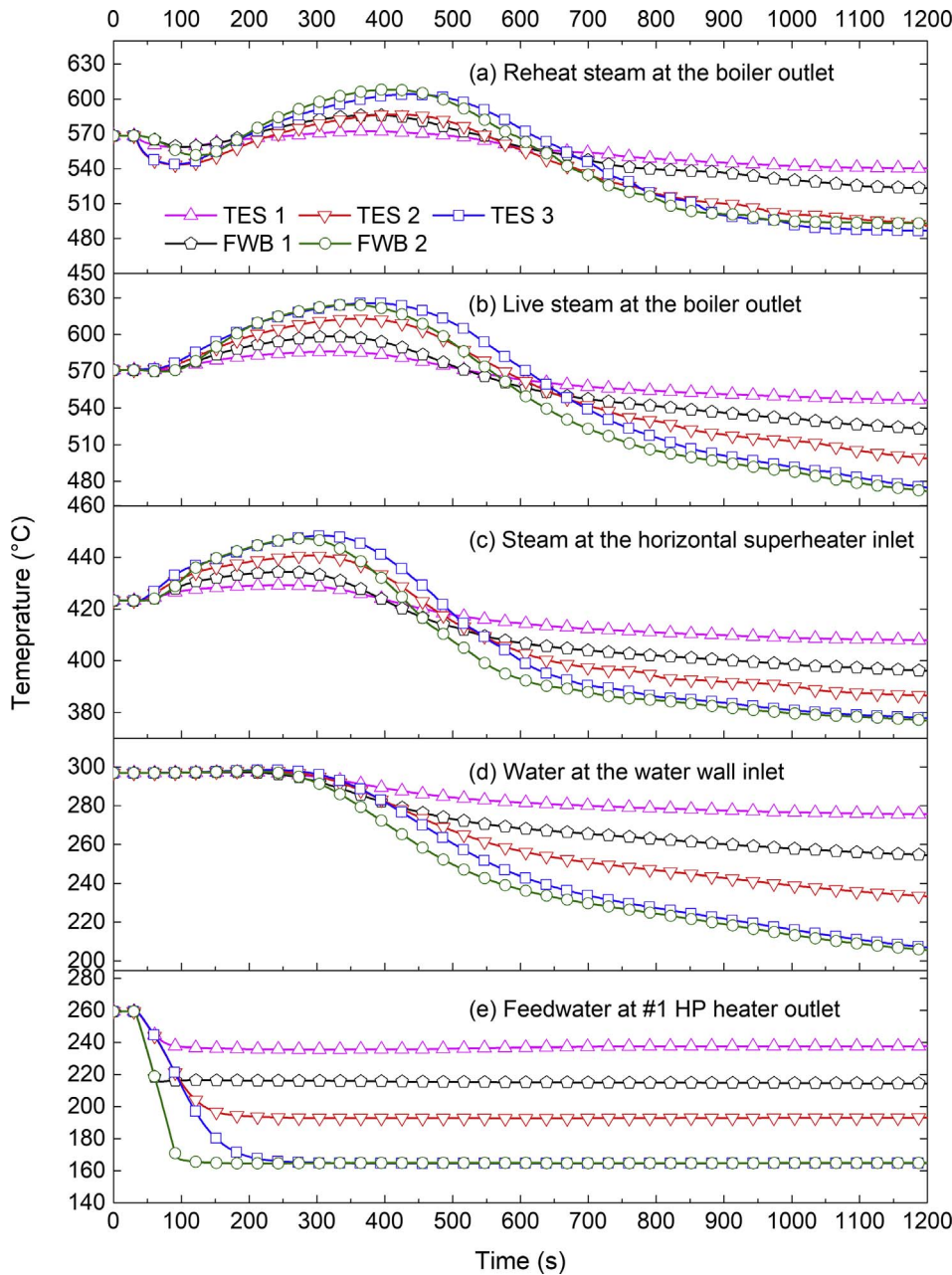


Fig. 9. Temperature dynamic responses of key parameters under five different measures.

extraction steam of #1, #2, and #3 HP heater) are nearly identical to those via FWB 2 (feedwater bypass 100%).

4.1.2. Dynamic responses of flowrate

The flowrate dynamic responses of the key parameters under five different measures are shown in Fig. 8. The results reveal that when the different measures are performed, the flowrates of reheat steam at the IP turbine inlet and exhaust steam are in agreement with the corresponding pressures. As the flowrate of feedwater that originated from the deaerator is fixed, the flowrate of feedwater at #1 heater outlet decreases sharply and then recovers to the initial value via TES measures, indicating that the feedwater is first stored and then released into the feedwater pipelines. However, the flowrate of feedwater increases sharply and then recovers to the initial value via FWB measures, indicating that the feedwater is first released and then stored into the feedwater bypass pipelines. The flowrate of live steam at the HP turbine inlet decreases sharply and then gradually recovers to the initial values. Some fluctuations are observed in the flowrate of water at the water

wall inlet because of the steam-liquid phase transformation in the water wall. Moreover, among the three TES measures, the change degrees of flowrate increase with the increment in the numbers of throttled valves. Between the two FWB measures, the change degrees of flowrate increase with the rise in the degree of feedwater bypass.

4.1.3. Dynamic responses of temperature

The temperature dynamic responses of the key parameters under five different measures are shown in Fig. 9. The results reveal that when the different measures are conducted, the temperature of feedwater at #1 HP heater outlet first responds, and the response rates of the temperature via TES measures are slower than those via FWB measures because the heat transfer in HP heaters is slower than flow propagation. The water temperature at the water wall inlet is maintained and then decreases rapidly because of the deficiency of thermal storage in the economizer system. Meanwhile, the live steam temperature at the boiler outlet and the steam temperature at the horizontal superheater inlet first increase to reach their peaks due to the decrement of steam

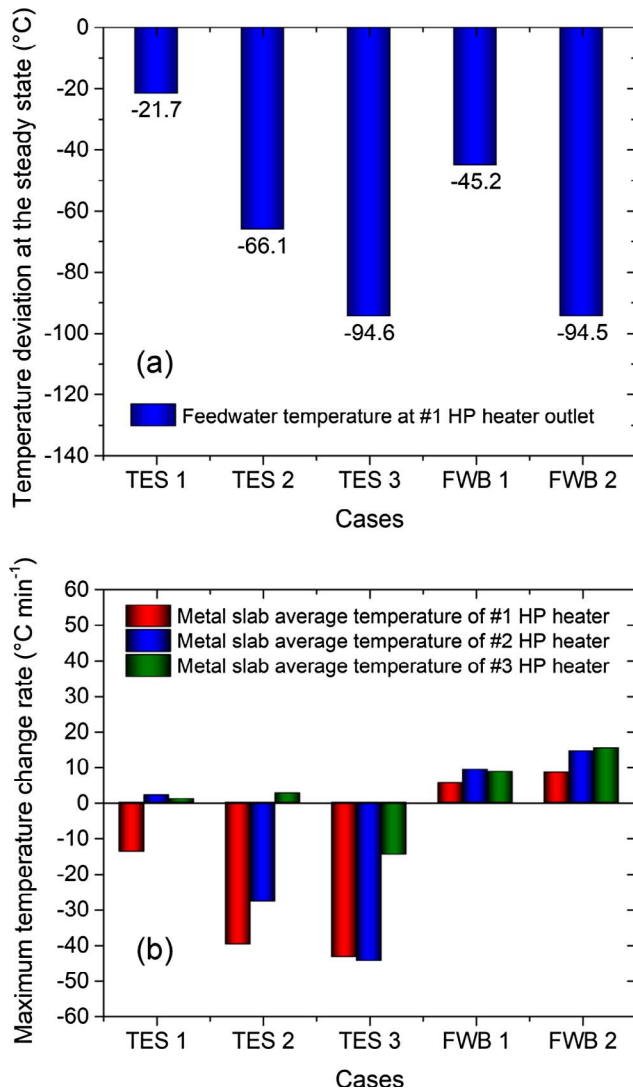


Fig. 10. Temperature change characteristics. (a) Feedwater at #1 HP heater outlet; (b) metal slabs in #1, #2, and #3 HP heaters under five different measures.

Table 4

Data of maximum temperature change rate for metal slabs in #1, #2 and #3 HP heaters under five different measures.

Measures	Maximum temperature change rate of metal slab (°C min ⁻¹)		
	In #1 HP heater	In #2 HP heater	In #3 HP heater
TES 1	-13.7	2.2	1.1
TES 2	-39.8	-27.8	2.7
TES 3	-43.5	-44.6	-14.5
FWB 1	5.5	9.3	8.7
FWB 2	8.6	14.5	15.3

flowrate, and then the temperatures decrease gradually because of the deficiency of thermal storage in the boiler system. The reheat steam temperature at the boiler outlet decreases sharply due to the increment of the reheat steam flowrate, and then the temperature reaches a plateau, and finally decreases gradually. Moreover, among the three TES measures, the change degrees of temperature increase with the increment in the number of throttled valves. Between the two FWB measures, the change degrees of temperature increase with the rise in the degree of feedwater bypass. Furthermore, on the steady state, the temperature of reheat steam, live steam, steam at the horizontal superheater inlet, water at the water wall inlet, and feedwater at #1 HP

heater outlet via TES 3 are all nearly identical to those via FWB 2.

The temperature deviations of feedwater at #1 HP heater outlet under five different measures are shown in Fig. 10(a). The results show that when compared with the initial value, the feedwater temperature at the steady state decreases by 21.7 °C, 66.1 °C, and 94.6 °C from TES 1 to TES 3, and decreases by 45.2 °C and 94.5 °C in FWB 1 and FWB 2, respectively. Although regulating extraction steam of HP heaters influences the temperature at the economizer inlet and finally results in larger exergy destruction in the boiler, increasing the share of intermittent renewable energy sources (RES) dramatically in power system poses challenges in maintaining grid stability and security in the present situation of China. Thus, confronted with the conflict between the safety of the power system and the economy of coal-fired power plant, the safety must be placed into priority, then the economy may be considered. Meanwhile, certain methods, such as adding heat tanks before the economizer and heating the feedwater using other renewable heat sources, can be used to improve the economy when the extraction steam of HP heaters is regulated. A feasibility study on these measures will be further conducted by our group. Moreover, regulating extraction steam of HP heaters quickly mainly influences the safety of HP heaters. Thus, the maximum temperature change rate for the metal slabs in #1, #2 and #3 HP heaters under five different measures are depicted in Fig. 10 (b) and listed in Table 4. The results show that among the direct measures (TES 1 to TES 3), the maximum temperature change rates of metal slabs increase with the increment in the number of throttled valves, and the maximum value of temperature change rate is -44.6 °C min⁻¹ occurred on the metal slab of the #2 HP heater at the TES 3 measures. Between the indirect measures (FWB 1 to FWB 2), the maximum temperature change rates of metal slabs increase with the rise in the degree of feedwater bypass, and the maximum value of temperature change rate is 15.3 °C min⁻¹ occurred on the metal slab of the #3 HP heater at the FWB 2 measures. It can be seen that the heaters are cooled quickly by throttling the extraction steams at the extraction steam pipelines, and a slight temperature increment occurred in the metal slabs of the HP heaters through feedwater bypass. Therefore, the safety of the heaters could be guaranteed.

4.2. Power dynamic responses and energy conversion

The dynamic responses of output power under five different measures are depicted in Fig. 11. The results show that when the different measures are performed, the output power undergoes two obvious stages. At the first stage, the output power increases sharply to reach a peak. At the second stage, the output power first decreases slowly to reach a plateau and then decreases rapidly. Finally, the output power becomes lower than the initial value. Similarly, among the three TES measures, the change degrees of power increase with the increment in the number of throttled valve, and between the two FWB measures, the change degrees of power increase with the rise in the degree of feedwater bypass.

Moreover, Fig. 6 shows the regulation of operational flexibility for coal-fired power plant. Two processes are included in the regulation, namely, power-up regulation and power-down regulation. Power-down regulation can be easily realized by throttling the HP governing valve, whereas the power-up regulation is more difficult to achieve, which is attributed to the shortage of thermal storage and time delay for heat transfer in the boiler system. In other words, regulating extraction steam of HP heaters and condensate throttling in the turbine system that release and use the thermal storage are the potential measures to compensate for the deficiency of the boiler system in the power-up regulation. Once the power-up regulation is finished in a short time, the measures in the turbine system can be regulated to the initial state gradually. Therefore, the power-up regulation process is under consideration.

Taking the power trend in the TES 2 as an example, Fig. 12 depicts the transfer and conversion process of energy to elucidate the power

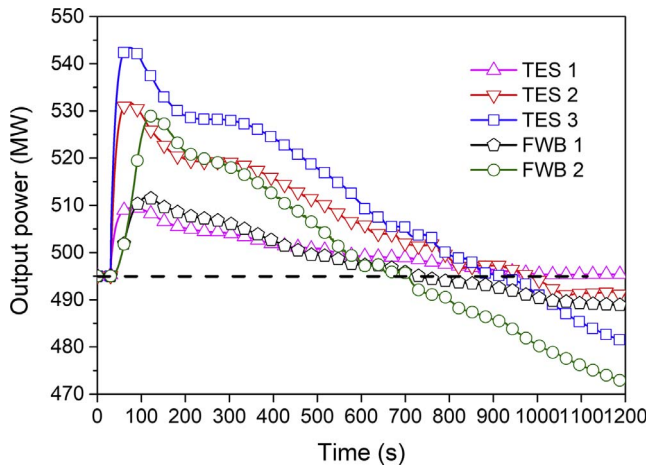


Fig. 11. Dynamic responses of output powers under five different measures.

trend. The results show that at stage 1, the extraction steam of HP heaters passes through the HP turbine, and the output power response rate is rapid due to the high velocity of steam in the turbine. Subsequently, the steam goes through the reheater, IP turbine, crossover, and LP turbine; thus, the flowrate of the exhaust steam is selected for the end signal of the first stage. The thermal storage in the boiler system and HP heater subsystem are the energy sources of the dynamic process, but the response rate of thermal storage is slow. Thus, the allotted time for stage 2 is long. Moreover, the activation of the thermal storage in the system slowly decreases the output power first to reach a plateau at the beginning of stage 2, and then the output power decreases quickly. Furthermore, after 985 s, the output power becomes less than the initial power because some thermal storages in the system have been used to produce additional output power, and the residual thermal storages are inadequate to produce the initial power unless more coal is added.

4.3. Operational flexibility of power system under the different measures

Operational flexibility refers to the capability of coal-fired power plant means to provide primary and secondary frequency control to stabilize the electrical grid. Therefore, the power ramp rate, power capacity, and energy capacity are the key metrics to evaluate the operational flexibility of the power system. In our study, under the five different regulating measures, the changes of these operational flexibility metrics are shown in Fig. 13 and listed in Table 5. The results reveal that the maximum value for the average power ramp rate in a minute is $40.89 \text{ MW min}^{-1}$ in TES 3, indicating that if valve 1, valve 2 and valve 3 are all cut off, then the additional 6.19% of rate power will be provided in the first minute. Meanwhile, the average power ramp rate is 4.68% of rate power per minute in TES 2, 3.06% of rate power per minute in FWB 2, 1.87% of rate power per minute in TES 1, and 1.06% of rate power per minute in FWB 1. Compared with all extraction steam throttling 100% (TES 3), the power ramp rate of feedwater bypass 100% (FWB 2) is lower because of longer regulating time and heat transfer delay between the extraction steam and feedwater in HP heater systems.

The power capacities in TES 3, TES 2, FWB 2, FWB 1, and TES 1 are 48.40, 36.78, 33.92, 16.72, and 14.53 MW, and the required times are 45, 43, 93, 81, and 51 s, respectively. As described, power capacities increase with the increment in the number of throttled valves and/or the degree of feedwater bypass. Among the three TES measures, the required times are almost identical due to the high velocity of steam. Between the two FWB measures, the required times are almost identical. Therefore, the regulation of the extraction steam of HP heaters is suitable for primary frequency control, and the potential of primary frequency control via TES measures is greater than those via FWB measures. Hu et al. [34] proposed a model of condensate throttling system for rapid load adjustment on the basis of the characteristics of turbine-stored energy on a 660 MW supercritical coal-fired power plant at 500 MW load working condition, which the condensate water flow

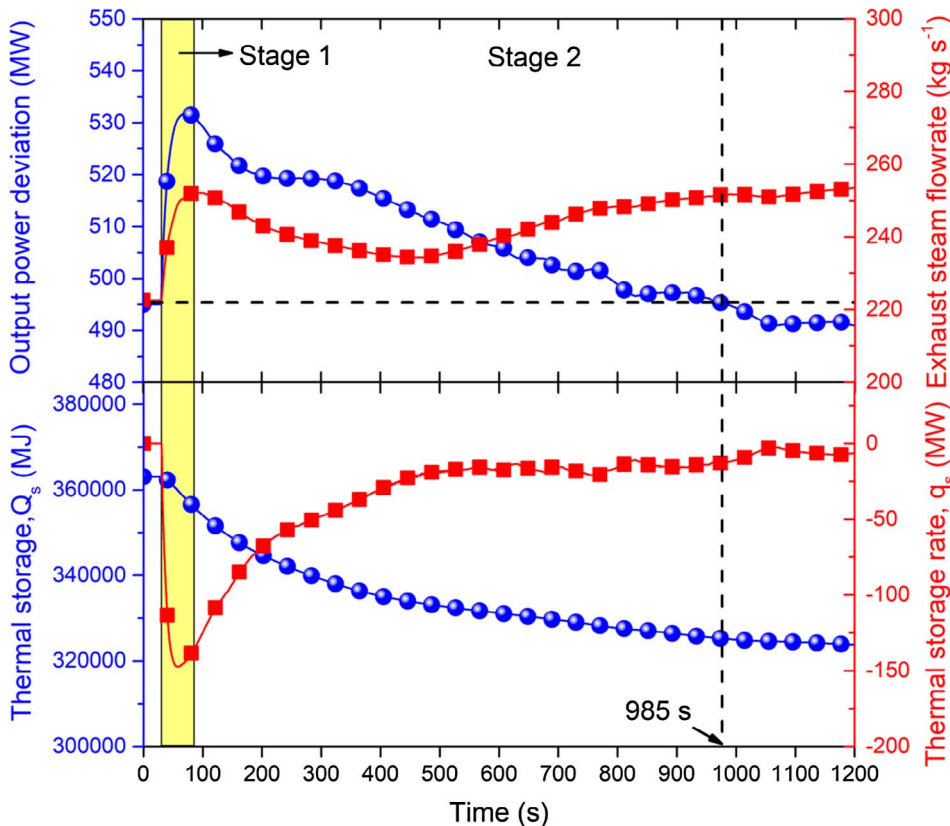


Fig. 12. Energy transfer and conversion process under TES 2 measure.

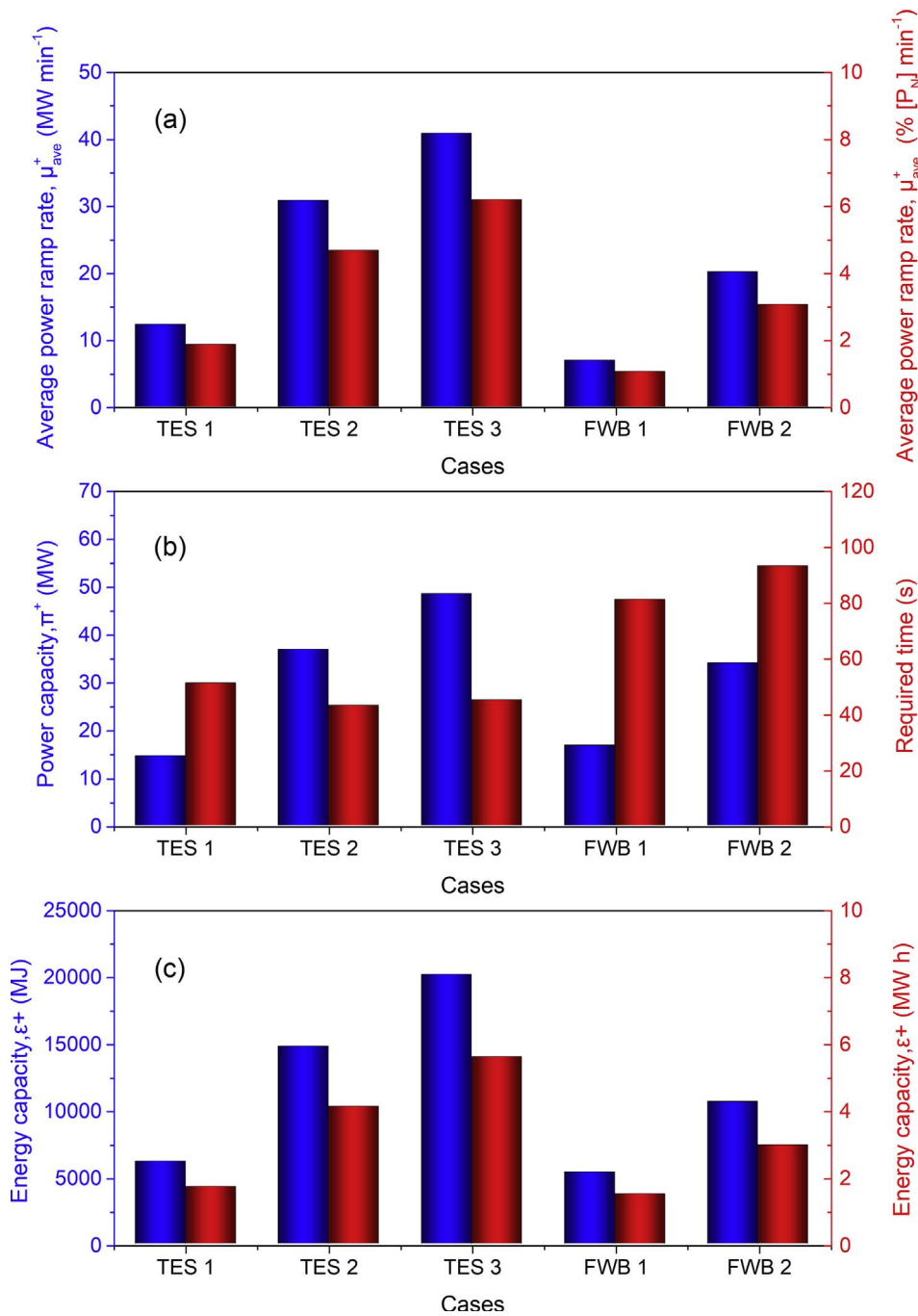


Fig. 13. Operational flexibility metrics of power system under five distinct measures. (a) Average power ramp rate within 1 min; (b) power capacity and required time; (c) energy capacity.

Table 5
Data of operational flexibility metrics for power system under five different measures.

Measures	Average power ramp rate MW min ⁻¹ ([P _N]% min ⁻¹)	Power capacity MW ([P _N]%)	Required time (s)	Energy capacity MJ (MW h)
TES 1	12.35 (1.87)	14.53 (2.20)	51	6147.14 (1.71)
TES 2	30.86 (4.68)	36.78 (5.57)	43	14747.68 (4.10)
TES 3	40.89 (6.19)	48.40 (7.33)	45	20098.18 (5.58)
FWB 1	6.99 (1.06)	16.72 (2.53)	81	5344.47 (1.48)
FWB 2	20.21 (3.06)	33.92 (5.14)	93	10608.78 (2.95)

decreases from 404 to 210 kg s⁻¹, and the unit load increases by approximately 7.4 MW. In our model, throttling the extraction steam of #1 HP heater (TES 1) is the worst measure for primary frequency control, and the power capacity is 14.53 MW, indicating that the regulating the extraction steam of HP heater is more effective for primary frequency control than condensate throttling.

Furthermore, energy capacity indicates the effective additional output power derived from the activated thermal storage in the specific thermal power unit. The energy capacities in TES 3, TES 2, FWB 2, TES 1, and FWB 1 are 5.58, 4.10, 2.95, 1.71, and 1.48 MW h, respectively. To conclude, regulating the extraction steam of HP heaters is also appropriate for secondary frequency control, and the potential of secondary frequency control varies according to different measures. To evaluate the operational flexibility of a specific thermal power plant more accurately and reliably, the addition of control strategies and

influencing factors merits further research.

5. Conclusion

The operational flexibility of a thermal power plant is a method to modulate the electrical power grid and compensate for the limitations of the renewable energy sources and load. Focusing on the regulation of the extraction steam of HP heaters applied in operational flexibility, dynamic simulations of an entire 660 MW supercritical coal-fired power plant were developed through GSE software, and the models were validated in the steady state and transient processes. Subsequently, five different measures were introduced, the dynamic response of the main thermodynamic parameters and output power were described, and the operational flexibility of these measures was discussed. The main conclusions are as follows:

- (1) For the main thermodynamic parameters, such as pressure, flowrate, and temperature, the response rates of pressure and flowrate are more rapid than temperature. Among the three TES measures, the change degrees of these parameters increase with the increment in the number of throttled valves. Furthermore, between the two FWB measures, the change degrees of those parameters increase with the rise in the degree of feedwater bypass. Compared with the initial value, the feedwater temperature at the steady state decreases by 21.7 °C, 66.1 °C, and 94.6 °C from TES 1 to TES 3, and decreases by 45.2 °C and 94.5 °C in FWB 1 and FWB 2, respectively. The maximum temperature change rate of metal slabs is $-44.6\text{ }^{\circ}\text{C min}^{-1}$ occurred on the metal slab of #2 HP heater at TES 3 measures.
- (2) Two ramp stages for the output power are observed. At the first stage, the extraction steam of the HP heater rapidly passes through the HP turbine, and the output power increases sharply to reach a peak. At the second stage, the thermal storage in the feedwater of HP heaters and the boiler system are activated, and the output power first decreases slowly to reach a plateau and then decreases rapidly. Finally, the output power becomes lower than the initial value.
- (3) The maximum average power ramp rate in a minute is 6.19% of rate power per minute (40.89 MW min^{-1}) in TES 3, and the minimum is 1.06% of rate power per minute in FWB 1. The maximum power capacity is 48.40 MW in TES 3, and the minimum is 14.53 MW in TES 1. Meanwhile, the required times by FWB measures are longer than those in TES measures. Furthermore, the maximum energy capacity is 5.58 MW h in TES 3, and the minimum is 1.48 MW h in FWB 1. The average power ramp rate, power capacity, and energy capacity increase with the increment in the number of throttled valves and/or the degree of feedwater bypass. Consequently, regulating the extraction steam of HP heaters is an effective measure to be applied in operational flexibility, including primary and secondary frequency control, and the potential for operational flexibility varies according to the different measures.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.apenergy.2018.01.017>.

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